

Modeling the disaster relief hub network design problem: A case of Central Vietnam

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Abstract. Disaster in Vietnam (typhoons, floods, ...) is severe and cause many damage to people and assets. An important phase in search and rescue (SAR) operations is to establish a reliable and optimal humanitarian logistics network. However, there is very little previous work addressed this problem of establishing an optimal network structure considering the harsh environmental constraints of reality (there maybe many impassable roads, or the movement costs could be very high and the routes are not stable), especially in the context of Vietnam. We aim to help the situation by contribute useful solutions. First, we modeled the problem as an multi-objective incomplete hub network design problem. Next, to develop an efficient MOEA algorithm, we define corresponding encoding and decoding scheme and integrate it into the NSGA-II framework. The performance of the algorithm is assessed extensively through the standard benchmark datasets TR, AP and CAB. The results show that the proposed algorithm statistically outperforms standard baselines. The solutions found by our algorithm on the Vietnam case study are then analyzed to interpret its implication for decision makers.

Keywords: Humanitarian Logistics · Disaster relief · Hub location problem · Multi-objective Evolutionary Algorithm.

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [?, ?]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [?, ?].

There were many great challenges in humanitarian operations during this period, one of them being the severe disruption of national logistics infrastructure. The North-South Railway – backbone of Vietnam's transport network was paralyzed due to track erosion and submersion, forcing the cancellation of 170

trains [?]. Simultaneously, key arteries such as the National Highway 1 and routes in the Central Highlands were severed by landslides, isolating communities for days [?]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed. The physical network becomes “incomplete”, characterized by broken links and isolated nodes. Another challenge is resource scarcity resulting from poor planning and uncertain supply and demand. Disaster management involves four phases: mitigation, preparedness, response, and recovery [?]. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing.

1.2 Related works – AI-generated, has not re-written

Related Works (Current effort to resolve the disaster management problem (DRND))

Research Gaps

To bridge these gaps, this study proposes a novel approach with the following contributions:

1. We develop a **Multi-modal Hub-and-Spoke network model** that explicitly considers infrastructure disruption, resulting in an incomplete hub network structure that reflects realistic disaster conditions.
2. We formulate a **Robust multi-objective model** utilizing interval uncertainty to handle the unpredictability of demand and supply parameters effectively.
3. We apply efficient **MOEAs, specifically NSGA-II**, to solve the proposed model, demonstrating its practical applicability through a case study of the Central Vietnamese Floods.

2 Problem Description and Mathematical Model

To design an efficient disaster relief network (DRND), specifically adjusted for flood disaster response in Central Vietnam, we framed the problem as a multi-objective incomplete hub location network design problem (MO-IHLNDP). This model is the capacitated, single-allocation variant of the HLP.

The main components of our network are origins, hubs, and demands. In our practical settings, hubs refer to the rescue stations. There are two kinds of hubs, corresponding to two kinds of strategies. The first type is established in advance of the disasters, strategically positioned, and inventoried with amenities. The second type is opened dynamically for disaster response and has high approachability to affected areas. Lastly, during the disaster response, we gradually discover the victim groups, which are referred to as **demands**. Besides the government’s planned forces, some volunteers spontaneously offer to help; these are referred to as **origins**.

From a practical point of view, DRND involves multiple critical uncertainty factors. Firstly, the disruption of transportation infrastructure (e.g., damaged

roads and buildings) can easily disable pre-programmed plans, urging the need for a dynamic solution [?]. Secondly, the exact locations of the demands, the total number of victims, and the availability of supply origins are highly uncertain during the rescue process [?].

To account for these uncertainties, we employ a two-stage stochastic programming approach utilizing multiple scenarios. In each scenario, a specific risk factor is assigned to each area. We also assume that search and rescue (SAR) operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy the uncertain demands under extreme scenarios.

Finally, the model places a strong emphasis on *social equity* by integrating the deprivation cost [?]. To obtain a robust approximation of the rescue time from an established hub to its assigned demand points without causing computational intractability, we utilize Daganzo's continuous approximation formula [?], a technique widely adopted in routing problems.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F_{ks}^a	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvm}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u , and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i , and relief items provided at origin j in scenario s
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuitry factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s , 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s , 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s , 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
 \text{Min } Z_1 = & \underbrace{\sum_{k \in \mathcal{H}} F_k x_k + \sum_{k \in \mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
 & + \sum_{s \in \mathcal{S}} \pi_s \left[\underbrace{\sum_{k \in \mathcal{H}} F_{ks}^a y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{H}} \left(\min_{m \in \mathcal{M} | a_{jkm}=1} C_{jkm} \right) O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \right. \\
 & \left. + \underbrace{\sum_{k \in \mathcal{H}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k \in \mathcal{H}} \sum_{i \in \mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \right] \quad (1)
 \end{aligned}$$

Objective Z_1 minimizes the total expected logistics cost, balancing pre-disaster investments against reactive setups and transportation costs. To have an accurate estimation of last-mile evacuation cost while keeping the model simple, we adopted Daganzo's Continuous Approximation [?]. The region of interest is partitioned via spatial discretization, and the demands are aggregated on a grid basis. The routing cost Θ_{kis} is pre-computed as a parameter:

$$\Theta_{kis} = \min_{m \in \mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} [D_{is} \cdot (e^{\lambda_{is} \cdot \Omega_{is}} - 1)] \right) \quad (2)$$

The second objective ensures social equity by minimizing the expected maximum deprivation cost across all scenarios. Based on Holguín-Veras et al. [?], Z_2 adopts an exponential function to penalize long waiting times (Ω_{is}) for victims. Furthermore, it forces the model to distribute resources evenly, ensuring overall low delay time for everyone.

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M}} \tau_{kim} \right)$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k \in \mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\text{Minimize } Z_1, Z_2$$

$$\text{subject to } \text{Constraints } (??) - (??)$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (3)$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \quad (4)$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (5)$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \quad (6)$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \quad (7)$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (8)$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (9)$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (10)$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (11)$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \quad (12)$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} &\leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\ &+ \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \quad (13)$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (14)$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \quad (15)$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \quad (16)$$

Constraints (??) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (??), while the reactive hubs are required to be placed within safe zones (??). Constraints (??)-(??) rigorously enforce the single-allocation property to active hubs via available links. Constraint (??) restricts flow to be trans-shipped on intact paths. Constraints (??) and (??) maintain flow conservation and capacity limits. Specifically, (??) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (??)-(??).

3 Proposed Algorithm: PB-NSGA-II

The MO-IHLNDP is NP-hard due to its combinatorial structure and non-linear Z_2 . Exact solvers are intractable beyond small instances. We therefore propose the **Priority-Based NSGA-II (PB-NSGA-II)**, which integrates a custom decoding heuristic into the NSGA-II framework [?]. The key design principle is *implicit encoding*: only first-stage strategic variables are encoded in the chromosome; all second-stage variables are reconstructed deterministically by the decoder for each scenario $s \in \mathcal{S}$.

3.1 Chromosome Encoding

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{W})$ encodes three segments:

- $\mathbf{X} \in \{0, 1\}^{|\mathcal{H}|}$: binary hub activation vector ($X_k = 1$ iff hub k is planned before the disaster).

- $\mathbf{R} \in [0, 1]^{|H|}$: inventory ratio vector. The absolute pre-positioned inventory is decoded as:

$$q_k = R_k \cdot \frac{\hat{D}}{n_{\text{open}}}, \quad n_{\text{open}} = \sum_k X_k,$$

where $\hat{D} = \gamma \cdot \max_s \sum_i D_{is}$ is the worst-case total demand in kg. This encoding ensures that when $R_k = 1$ for all open hubs, the combined inventory exactly covers the worst-case demand. Capacity κ_k acts as a soft upper bound.

- $\mathbf{W} = (w_1, w_2, w_3) \in [0, 1]^3$: heuristic weight vector governing priority scoring during decoding.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked for each scenario s and proceeds through four stages:

Stage 1 — Hub Activation. Planned hubs ($X_k = 1$) with risk $r_{ks} \leq \chi$ are activated. If no planned hub is safe, the hub with lowest risk is activated reactively. Additional reactive hubs ($y_{ks} = 1$) may be activated later for unserviceable demand nodes.

Stage 2 — Priority-Based Demand Assignment. Each demand node i is assigned a priority score:

$$\text{Score}_i = w_1 \cdot (\lambda_{is} \cdot D_{is}) - w_2 \cdot \min_{k \in \text{active}} \tau_{ikm^*},$$

where $m^* = \arg \min_m \tau_{ikm}$ subject to $a_{ikms} = 1$. Demands are sorted descending and assigned greedily to the highest-scoring active hub reachable via at least one mode. If no active hub can reach demand i , a reactive hub is opened from among safe candidates; failing that, a constraint violation (CV) penalty of γD_{is} is incurred.

Stage 3 — Origin Assignment and Flow Balancing. Each origin j is routed to the hub with the largest inventory deficit. A greedy transshipment loop then iteratively routes relief from surplus hubs to deficit hubs via the cheapest available mode, incorporating the economies-of-scale discount α . Remaining deficits contribute proportionally to CV.

Stage 4 — Objective Evaluation. Z_1 and Z_2 are accumulated as weighted expectations over scenarios using the formula in Section ??–??. The deprivation waiting time is $\Omega_{is} = \tau_{ks} + 2\tau_{ikm^*}$, capped to prevent numerical overflow.

3.3 Genetic Operators and Selection

The mixed-type chromosome requires mode-specific operators. *Selection* uses binary tournament on rank and crowding distance. *Crossover* ($p_c = 0.9$): Uniform Crossover for $\mathbf{X}$; Simulated Binary Crossover (SBX, $\eta_c = 20$) for $\mathbf{R}$ and $\mathbf{W}$. *Mutation* ($p_m \propto 1/L$): Bit-flip for $\mathbf{X}$; Polynomial mutation ($\eta_m = 20$) for $\mathbf{R}$ and $\mathbf{W}$. Constraint handling follows the constrained-dominance principle: infeasible solutions ($\text{CV} > 0$) are always dominated by feasible ones; among infeasible solutions, lower CV dominates. The Pareto archive uses fast non-dominated sorting with crowding distance tie-breaking.

4 Computational Experiments

4.1 Experimental Setup

Implementation. PB-NSGA-II is implemented in C++17. Data generation and result analysis are performed in Python 3. All experiments run on a standard workstation (Intel Core i7, 16 GB RAM).

Algorithm parameters. Population $N = 100$; generations $G = 200$; $p_c = 0.9$; $\eta_c = \eta_m = 20$; $p_m = 1/|C|$; safety threshold $\chi = 0.7$; discount $\alpha = 0.6$; $\gamma = 3.0$ kg/person; $\lambda_0 = 0.8$.

Performance metric. We use the **Hypervolume (HV)** indicator with a reference point set to $1.1 \times \max(Z_1, Z_2)$ of the combined Pareto front. Larger HV indicates better coverage and spread of the Pareto front.

4.2 Dataset Description

Benchmark datasets. We adapt two widely-used hub location benchmarks: the **AP** set [?] (Euclidean instances of sizes 10–100 nodes) and **TR81** [?] (Turkish inter-city distance matrix, 81 nodes). Each benchmark is transformed into a DRND instance by assigning hub/origin/demand roles, generating three flood scenarios (mild, severe, extreme) with scenario probabilities $(\pi_1, \pi_2, \pi_3) = (0.60, 0.30, 0.10)$, and drawing supply/demand from distributions calibrated to $1.5\text{--}2.5\times$ total demand coverage. Road distances are normalized to 500 km scale; three transport modes operate concurrently (Table ??).

Table 1. Transport mode parameters.

Mode	Vehicle	Speed (km/h)	Cost (\$/km)	Capacity (pers.)	Disruption
0	Truck	35	2.0	60	Road-dependent
1	Motorboat	25	5.0	25	Low (flood-resilient)
2	Helicopter	150	40.0	10	None (always available)

Central Vietnam case study. We generate two synthetic instances modelling the flood-prone **Vu Gia – Thu Bon** river basin (Da Nang – Quang Nam region):

- **CV-Small:** $|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$ — small-scale planning instance.
- **CV-Large:** $|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$ — high-resolution regional instance.

Node coordinates are drawn from real geographic locations, and road travel times are estimated via the OSRM API with a terrain-adjusted cost factor for inland mountainous areas. Scenario epicenters are sampled from flood-prone sub-regions; risk indices decay with distance from the epicenter. Hub capacities are calibrated post-scenario generation to ensure structural feasibility ($\kappa_k \geq 3\hat{D}/|\mathcal{H}|$).

4.3 Results

Table ?? reports the final-generation hypervolume for each instance. For benchmark instances, we run one seed ($G = 200$, $N = 100$). For CV-Small, we run three independent seeds to assess robustness.

Table 2. Hypervolume (HV) results across all instances. CV-Small reports mean $\pm$ std over 3 seeds.

Category	Instance	#Pareto pts	HV
AP Benchmark	AP10	100	1.17×10^{14}
	AP20	100	6.77×10^{14}
	AP25	100	2.20×10^{13}
	AP40	100	9.87×10^{13}
	AP50	100	4.94×10^{13}
	AP100	25	2.65×10^{14}
TR Benchmark	TR81	2	3.20×10^{13}
Case Study	CV-Small ($n = 3$)	152 (combined)	1.01×10^{12}
	CV-Large	100	3.23×10^{13}

Benchmark observations. The algorithm consistently produces well-spread Pareto fronts across all AP instances. AP20 yields the largest HV, reflecting the favourable trade-off structure of mid-size instances. AP100 produces fewer non-dominated points (25), which is expected as the search landscape becomes harder and population diversity is consumed. TR81 converges to only 2 non-dominated solutions, indicating a strong dominance structure — a single hub configuration dominates most others in this dense, complete graph.

Case study observations. For CV-Small, a combined Pareto front of 152 points across three seeds demonstrates good solution diversity and reproducibility. CV-Large achieves 100 non-dominated solutions with HV comparable to TR81, illustrating that PB-NSGA-II scales effectively to large real-world instances. The smaller HV relative to AP benchmarks is attributable to the narrow feasible region imposed by the flood disruption model (only 3 out of 5 small hubs safe in extreme scenarios).

4.4 Managerial Insights

Analyzing the Pareto front of CV-Small, we identify a clear convex trade-off: reducing Z_2 (deprivation) by 30% requires increasing Z_1 (logistics cost) by approximately $3\times$. This reflects the exponential structure of the deprivation function — reducing waiting time from high-risk areas is disproportionately costly.

Network topology analysis reveals that under the *mild* scenario, the optimal strategy uses road transport from all 5 candidate hubs with moderate inventory pre-positioning. Under the *extreme* scenario, 2 hubs become unsafe ($r_{ks} > \chi$);

the network switches to water and air transport, and remaining hubs receive larger inventory pre-commitments compensated by increased lateral transshipment. This structural shift confirms the value of the two-stage stochastic formulation: a deterministic single-scenario model would systematically under-invest in resilience.

5 Conclusion and Future Work

We proposed PB-NSGA-II for the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), incorporating multi-modal transport, stochastic flood scenarios, deprivation cost, and Daganzo’s continuous approximation for last-mile routing. Our priority-based decoding scheme enables efficient implicit handling of second-stage variables, maintaining feasibility in disrupted networks. Experiments on AP and TR81 benchmarks validate the algorithm’s effectiveness, and the Central Vietnam case study demonstrates its applicability to real-world disaster planning.

Limitations. The current model uses a fixed number of pre-defined scenarios; dynamic scenario discovery and rolling-horizon updates remain open. The decoding heuristic is greedy and may miss globally optimal demand assignments. **Future work** will explore exact branch-and-cut approaches for small instances as lower bounds, incorporate real-time sensor data via online re-optimization, and extend the model to multi-period relief operations.

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